ANALYSIS REPORT

Wave 5.2R Stage 7 Mean And Centered-Shape Multi-Head Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Decision

Stage 7 is complete as a valid negative result.

All `7 / 7` first-screen runs completed without failure. No shared or partially shared candidate passed the complete multi-index gate, so the conditional stability continuation was correctly skipped.

C01, the monolithic H04 fine-tuning control, is the raw-error leader at `0.001712731 deg`. It improves raw MAE by `0.76%` and mean MAE by `2.32%` relative to frozen H04, but centered-shape MAE changes by `-0.04%`. C01 is not a multi-head candidate and does not qualify for promotion.

Stage 5 H04 remains the qualified structured component entering Stage 8. No Stage 7 model replaces the accepted periodic GRU or becomes a production candidate.

Scope And Integrity

- dataset: `polished_dataset`;
- input contract: setpoints only;
- surface: `Fw`;
- accepted curves: `966`;
- split: `675` train, `194` validation, `97` test;
- angular grid: `2048` uniform points;
- split signature: `c1aa8718fb9bf88cc2021c121dc4f3b4010fc1d2e45ac90af5f4376aa64f8e16`;
- first-screen seed: `314159`;
- completed runs: `7 / 7`;
- target-derived runtime inputs: none;
- official TE Curve Verification Pipeline: not run.

  Every candidate satisfies exact decomposition invariants. Maximum observed centered-shape mean is below `3.4e-10 deg`, and reconstruction identity error is exactly zero on the test split.

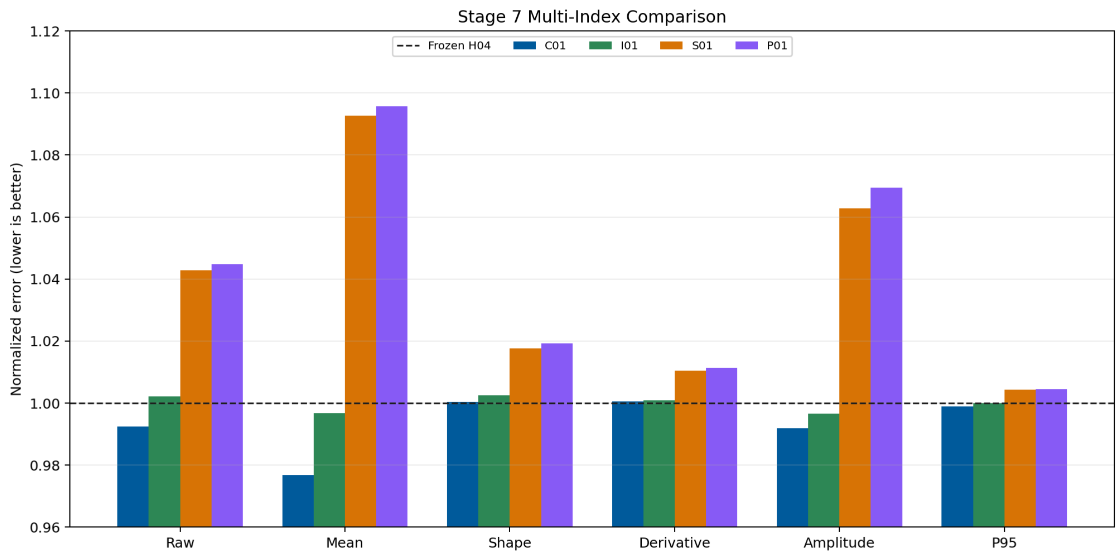
Candidate Matrix

ID	Architecture	Parameters	Relative to I01	Role
C01	monolithic H04	7123	0.523	fine-tuning control
S01	fully shared heads	8179	0.601	promotion candidate
P01	partially shared	10259	0.753	promotion candidate

ID	Architecture	Parameters	Relative to I01	Role
I01	independent heads	13619	1.000	matched control
G01	shared plus projection	8179	0.601	promotion candidate
A01	analytical mean	7090	0.521	ablation
A02	analytical shape	6529	0.479	ablation

Primary Results

ID	Raw [deg]	Mean [deg]	Shape [deg]	D-MAE	Phase [rad]	P95 [deg]
Frozen H04	0.0017259	0.0008844	0.0013555	0.2810657	0.3210167	0.0039334
C01	0.0017127	0.0008638	0.0013561	0.2812074	0.3141944	0.0039294
I01	0.0017296	0.0008816	0.0013589	0.2813380	0.3511679	0.0039332
S01	0.0017998	0.0009664	0.0013794	0.2839949	0.3298313	0.0039502
P01	0.0018031	0.0009690	0.0013817	0.2842617	0.3333249	0.0039511
G01	0.0017998	0.0009664	0.0013794	0.2839949	0.3298313	0.0039502



Stage 7 multi-index comparison

Gate Matrix

ID	Raw	Mean	Shape	Deriv.	Harm.	P95	Shared	Invariant	Final
S01	fail	fail	fail	fail	fail	fail	fail	pass	fail
P01	fail	fail	fail	fail	fail	fail	fail	pass	fail
G01	fail	fail	fail	fail	fail	fail	fail	pass	fail

No shared formulation passes any predictive improvement gate. The structural invariants pass for all candidates.

What Worked

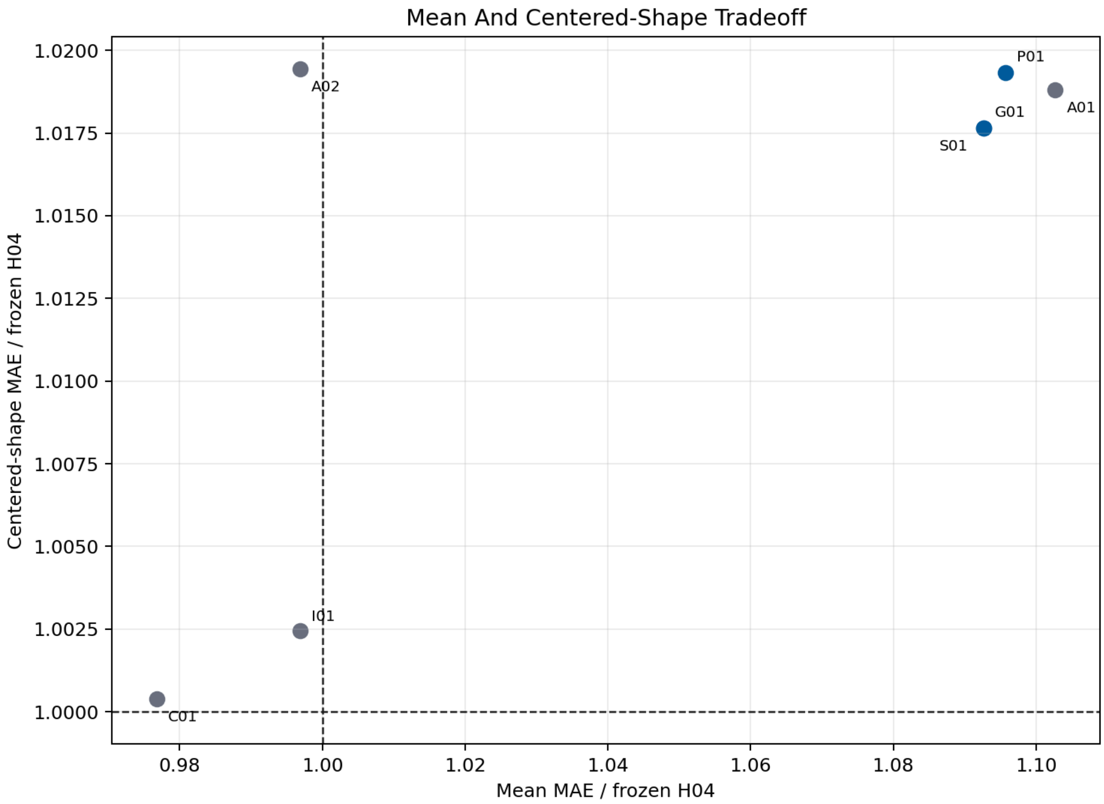
- the mean-plus-shape reconstruction is exact and numerically stable;
- S01 uses `60.1%` and P01 uses `75.3%` of I01 parameters;
- C01 shows that continued bounded H04 optimization can improve raw and mean error while preserving closure, amplitude, phase, and P95;
- the campaign directly measures mean-shape gradient conflict;
- the analytical ablations localize the need to learn both components.

What Did Not Work

- S01, P01, and G01 worsen raw, mean, shape, derivative, harmonic, and P95 behavior relative to frozen H04;
- the parameter savings do not compensate for their predictive regression;
- I01 is the closest balanced decomposition but does not achieve the required mean and shape gains despite using the most parameters;
- A01 shows that a frozen analytical mean is insufficient;
- A02 shows that learning only the mean does not preserve the qualified shape;
- no candidate earns stability continuation or promotion.

Mean And Shape Tradeoff

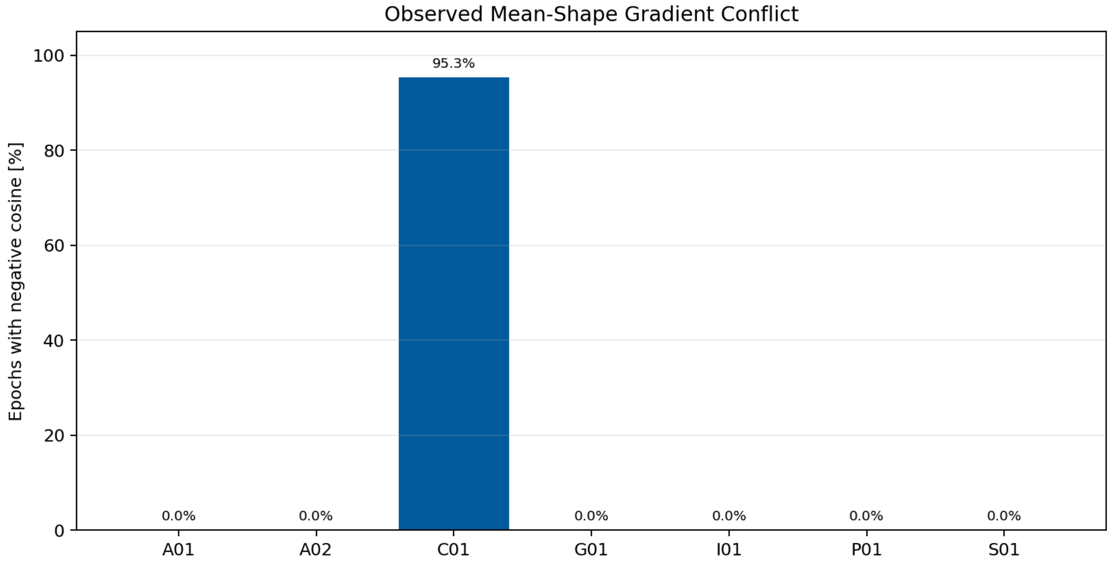
The lower-left quadrant improves both explicit quantities. C01 improves the mean but sits slightly above the frozen shape baseline. Every multi-head candidate remains outside the required improvement region.



Mean and shape tradeoff

Gradient Conflict

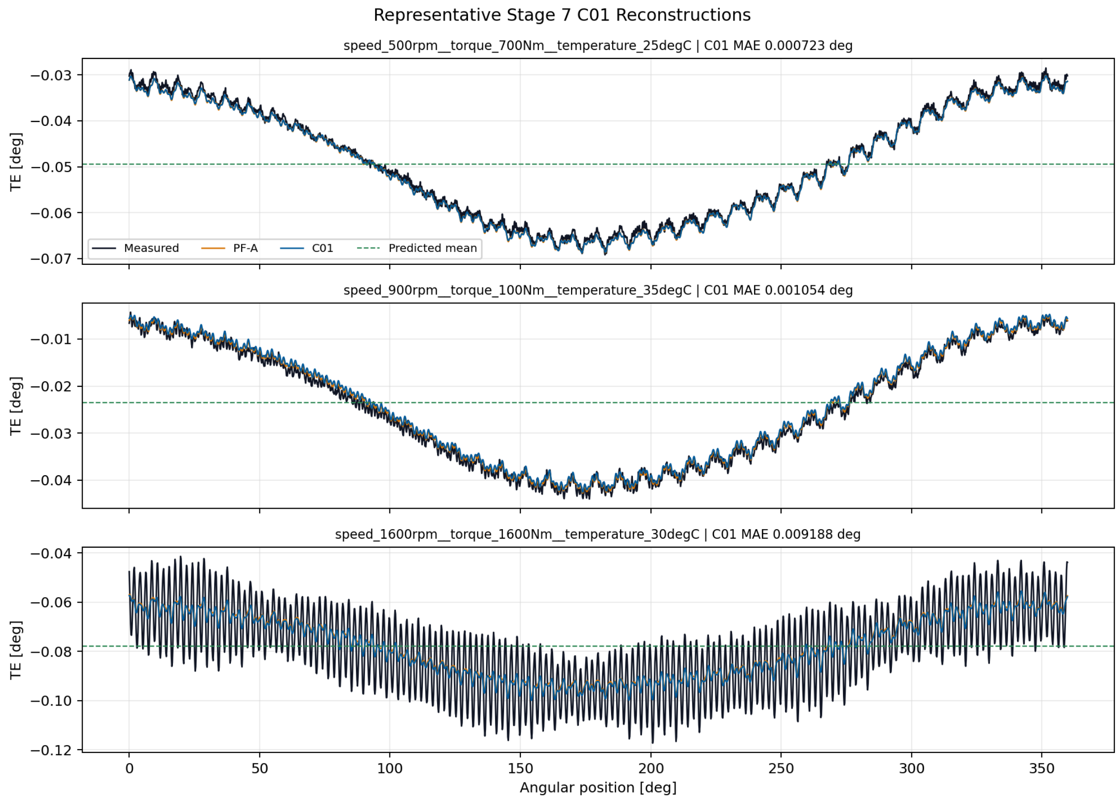
C01 records negative mean-versus-shape cosine in `95.3%` of epochs. In the explicit shared-head models the measured shared gradient cosine is non-negative, so G01's projection is never activated and G01 is numerically equivalent to S01. This explains why gradient surgery does not recover performance in this screen.



Mean-shape gradient conflict

Representative Full Curves

C01 remains close to frozen H04 and improves the mean component, but its worst cell still exposes unresolved high-order shape error.



Representative C01 curves

Scientific Interpretation

Exact decomposition improves interpretability but does not by itself add predictive information. The current dataset and bounded coefficient target allow the monolithic model to trade mean against small shape changes more effectively than the explicit multi-head models.

The result also distinguishes architectural conflict from gradient conflict. C01 exhibits frequent negative cosine, while explicit shared heads do not. Their failure therefore cannot be repaired by conflict projection alone; the shared representation and optimization path are the limiting factors in this screen.

Stage 8 returns to a weaker, mechanism-specific hypothesis: forward compliance priors. It will start with diagnostics and sign-only or broad-bound constraints rather than a hard equation.

Program Decision

- Stage 7 status: complete, valid negative result;
- completed runs: `7 / 7`;
- stability runs: `0`, correctly skipped;
- raw-error leader: C01 monolithic control;
- promoted Stage 7 candidate: none;
- retained component: Stage 5 H04;
- production or registry promotion: no;
- next step: Stage 8, Weak Forward Compliance Priors.

Artifact Map

- campaign: `output/training_campaigns/2026-07-29-17-46-21_wave52r_stage7_mean_centered_shape_multi_head_2026_07_29/`;
- gate summary: `output/analysis/wave_5_2r/stage7_mean_centered_shape_multi_head/closeout/stage7_exit_gate_summary.yaml`;
- preflight: `output/analysis/wave_5_2r/stage7_mean_centered_shape_multi_head/stage7_preflight_validation_summary.yaml`;
- C01 checkpoint: `output/training_runs/mean_centered_shape_multi_head/2026-07-29-17-46-25__stage7_c01/best_model.pt`;
- model report: `doc/reports/analysis/model_development_waves/wave_5_2/physics_guided_pinn_reassessment/[2026-07-29]/stage7_mean_centered_shape_multi_head/stage7_mean_centered_shape_multi_head_model_report.md`.